

Managing Applications with AWS Elastic Beanstalk

AWS Elastic Beanstalk provides a platform for deploying and managing applications (versions) in the AWS Cloud. This article explains how it works, and lists its key components as well as the platforms it supports.



There are multiple platforms (Java, .NET, Node.js, PHP, Go, Python, and Ruby) available that can be used to upload an application package file or store it in Amazon S3. AWS Elastic Beanstalk helps you to manage all the resources that are needed to run applications. It manages such features as load balancing, scaling, and monitoring.

Key components

Table 1 lists the key components of Elastic Beanstalk.

Component	Role
Application	Logical collection (folder) of Elastic Beanstalk components such as environments, versions, and environment configurations
Application version	Specific iteration of deployable code for an application. An application can have multiple versions
Environment	Collection of AWS resources that runs one application version at a time. However, one version can be deployed in multiple environments
Environment tier	Type of application that the environment runs, and provides resources accordingly; for example, Web server environment tier and worker environment tier
Environment configuration	Collection of parameters and settings -- load balancer, auto scaling group, database, security, and so on
Platform	Combination of an operating system, runtime environment, Web server/application server, and Elastic Beanstalk components

Table 1: Key components of Elastic Beanstalk

The screenshot shows the Amazon Elastic Beanstalk console. The main heading is 'Amazon Elastic Beanstalk End-to-end web application management.' Below this, there is a 'Get started' section with a 'Create Application' button. The 'How it works' section explains that you simply upload your code and Elastic Beanstalk handles the deployment, from capacity provisioning, load balancing, and automatic scaling to web application health monitoring, with ongoing fully managed patch and security updates. The 'Pricing' section states that there is no additional charge for Elastic Beanstalk. The 'Benefits and features' section is partially visible at the bottom.

Figure 1: Create an application